

Reinforcement Learning Laboratory

Lab Assignment 3: GridWorld – Policy Evaluation and Value Iteration

Task 1: Formulating the GridWorld Environment

RL Component	Description
States	S0, S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14 and G
Actions	Up, Down, Left, Right
Reward Structure	-1 for every normal move and +10 for reaching the goal G
Terminal State(s)	G
Goal of the Agent	Reach the goal state G using the minimum number of steps while maximizing cumulative reward

Task 2: Policy Evaluation

Using $\gamma = 0.9$ and starting with values of 0:

Iteration	Maximum Change (Δ)	Convergence Status
1	10.000	Not Converged
2	8.000	Not Converged
3	6.200	Not Converged
4	4.580	Not Converged
5	3.122	Not Converged

Task 3: Value Iteration and Optimal Policy Generation

Bellman Optimality Equation = $V_{k+1}(s) = \max_a [R(s,a) + \gamma V_k(s')]$

- $\gamma = 0.9$
- Normal move reward = -1
- Reaching G = +10
- G is terminal, so $V(G)=0$

$$V(S14)=10+0.9(0) = 10$$

$$V(S13)=-1+0.9(10) = -1+9 = 8$$

$$V(S12)=-1+0.9(8) = 6.2$$

So like that calculated for all cells.

State	Optimal Action	State Value
S0	Right / Down	1.810
S1	Right / Down	3.122
S2	Right / Down	4.580
S3	Down	6.200
S4	Right / Down	3.122
S5	Right / Down	4.580
S6	Right / Down	6.200
S7	Down	8.000
S8	Right / Down	4.580
S9	Right / Down	6.200
S10	Right / Down	8.000
S11	Down	10.000
S12	Right	6.200
S13	Right	8.000
S14	Right	10.000
G	Terminal	0.000

Task 4: Optimal Path Analysis

$S0 \rightarrow S1 \rightarrow S2 \rightarrow S3 \rightarrow S7 \rightarrow S11 \rightarrow G$

Total reward = $-1 -1 -1 -1 -1 + 10 = +5$

Policy Type	Path Length	Total Reward	Goal Reached (Yes/No)
Random Policy	Depends on random actions	Depends on path	Yes/No
Evaluated Policy	6	+5	Yes
Optimal Policy	6	+5	Yes

Analysis Questions

1. What is a state in GridWorld?

➔ A state represents the **current position of the agent in the GridWorld**. For example, S0, S1, S2, ..., S14 represent different positions, while G is the goal state.

2. What actions can the agent perform?

➔ The agent can perform four actions: **Up, Down, Left, and Right**.

3. What is the purpose of Policy Evaluation?

➔ The purpose of Policy Evaluation is to **calculate the value of each state for a given policy**. It shows how good it is for the agent to be in each state when following that policy.

4. How does Value Iteration differ from Policy Evaluation?

→ **Policy Evaluation** evaluates a fixed policy and calculates its state values. **Value Iteration** directly searches for the optimal state values by selecting the best possible action at each state and then derives the optimal policy.

5. What is the Bellman Optimality Equation?

→ The Bellman Optimality Equation gives the maximum expected value obtainable from a state:

$$V^*(s) = \max_a [R(s,a) + \gamma V^*(s')]]$$

It selects the action that gives the **highest expected return**.

6. Which policy reached the goal using the fewest steps?

→ The **Optimal Policy** reached the goal using the fewest steps because it selects the best action at every state.

7. Why is the optimal policy superior to a random policy?

→ The optimal policy is superior because it **chooses actions that maximize the expected reward and generally reaches the goal in fewer steps**. A random policy chooses actions without considering their consequences.

8. How would obstacles affect the optimal path?

→ Obstacles would prevent the agent from moving through certain states. The optimal policy would therefore **choose an alternative path around the obstacles**, which may increase the number of steps required to reach the goal.